

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

Register Number	192312443
Name	P.Ramesh

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

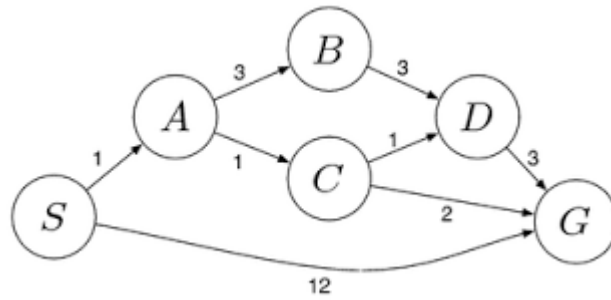
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path

from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I P.Ramesh(192312443) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Ramesh

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

1)ANS;

Problem

- A 4-gallon jug
- A 3-gallon jug
- A pump to fill the jugs

Goal: Obtain exactly 2 gallons of water in the 4-gallon jug.

Initial State

(0,0)

(4-gallon jug, 3-gallon jug)

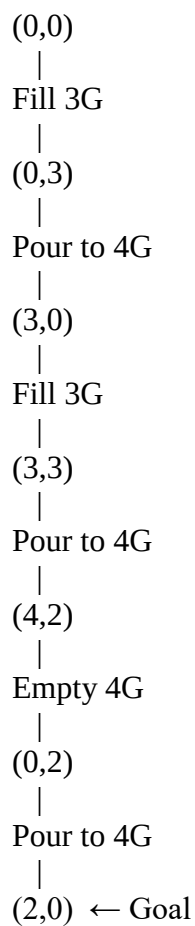
Goal State

(2, 0)

Solution step;

Step Action	State (4G, 3G)
1 Fill the 3-gallon jug	(0, 3)
2 Pour 3G into 4G	(3, 0)
3 Fill the 3-gallon jug again	(3, 3)
4 Pour from 3G to 4G until 4G is full	(4, 2)
5 Empty the 4-gallon jug	(0, 2)
6 Pour the remaining 2 gallons from 3G into 4G	(2, 0)

State Diagram



2)ANS;

1)

- Images from cameras
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Terrain and obstacle information

2)

- Partially Observable: The rover cannot see the entire environment at once.
- Sequential: Each action affects future actions.
- Dynamic: Weather and terrain may change over time.
- Continuous: Movement, time, and position change continuously.

3)

- Move forward, backward, left, and right.
- Capture images and videos.
- Collect rock and soil samples.
- Drill rocks for analysis.
- Analyse collected samples.

4)

- Accuracy of scientific data collected.
- Number of successful samples collected.
- Distance explored.
- Safe navigation without collisions.
- Battery and energy efficiency.

5)

- The rover has clear mission goals such as exploring Mars, collecting samples, and transmitting data.
- It plans actions to reach scientific targets safely.
- It selects the best path while avoiding obstacles and conserving energy.

3)ANS;

1. Initial State

- An empty 8×8 chessboard with no queens placed.

2. State Space

- All possible arrangements of queens on the chessboard.
- Each state represents a partial or complete placement of queens.

3. Operators (Actions)

- Place one queen in a safe position in the next column.
- If no safe position exists, backtrack and try another position.

4. Goal State

- Place all 8 queens on the board.
- No two queens share the same row, column, or diagonal.

5. Constraints

- One queen per row.
- One queen per column.
- No two queens on the same diagonal.

4)ANS;

- Initial State: Customer opens the OLA app.
- Goal State: Customer reaches the destination.
- Actions:
 - Enter source location.
 - Enter destination.
 - Confirm booking.
 - Make payment.
- Performance Measure:
 - Minimum waiting time.
 - Lowest fare.
 - Safe journey.

Pseudocode;

BEGIN

Input Source Location

Input Destination Location

Display Available Cab Types

(Mini, Micro, Sedan, Shared, Prime)

User selects Cab Type

Check Cab Availability

IF Cab is Available THEN

Calculate Fare

Display Driver Details

Confirm Booking

Assign Driver

Start Journey

Reach Destination

Make Payment

Display "Trip Completed Successfully"

ELSE

Display "Cab Not Available"

Ask User to Select Another Cab

ENDIF

END

5)ANS;

Given Weighted Graph

- $S \rightarrow A = 1$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$

- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

UCS Steps

Step	Expanded Node	Path Cost	Priority Queue
1	S	0	A (1)
2	A	1	C (2), B (4)
3	C	2	D (3), G (4), B (4)
4	D	3	G (4), B (4), G (6)
5	G	4	Goal Reached

Possible Paths

Path	Cost
$S \rightarrow A \rightarrow C \rightarrow G$	4
$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	6
$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$	10

Least-Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost = 4

